

Apply filters to SQL queries

Project description

My organization is working to make their system more secure. It is my job to ensure the system is safe, investigate all potential security issues and update employee computers as needed. The following steps provide examples of how I used SQL with filters to perform.

Retrieve after hours failed login attempts

There was a potential security incident that occurred after business hours (after 18:00). All after hours login attempts that failed need to be investigated.

The following code demonstrates how I created a SQL query to filter for failed login attempts that occurred after business hours.

```
MariaDB [organization]> SELECT * FROM log_in_attempts WHERE login_time > '18:00' AND success = 0;
```

	event_id	username	login_date	login_time	country	ip_address
205.12	2	apatel	2022-05-10	20:27:27	CAN	192.168.
66.142	18	pwashing	2022-05-11	19:28:50	US	192.168.
109.50	20	tshah	2022-05-12	18:56:36	MEXICO	192.168.
27.57	28	aestrada	2022-05-09	19:28:12	MEXICO	192.168.
45.93	34	drosas	2022-05-11	21:02:04	US	192.168.
4.157	42	cgriffin	2022-05-09	23:04:05	US	192.168.
58.57	52	cjackson	2022-05-10	22:07:07	CAN	192.168.
100.17	69	wjaffrey	2022-05-11	19:55:15	USA	192.168.
234.49	82	abernard	2022-05-12	23:38:46	MEX	192.168.
	87	apatel	2022-05-08	22:38:31	CANADA	192.168.

Filtered the login attempts which are after business hours and failed to login. First condition is `login_time > '18:00'`, which filters for the login attempts that occurred after 18:00. The second condition is `success = 0`, which filters for the failed login attempts.

Retrieve login attempts on specific dates

A suspicious event occurred on 2022-05-09. Any login activity that happened on 2022-05-09 or on the day before needs to be investigated.

```
MariaDB [organization]> SELECT * FROM log_in_attempts WHERE login_date = '2022-05-09' OR '2022-05-08';
```

	event_id	username	login_date	login_time	country	ip_address
	1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.1
40	1					
	2	apatel	2022-05-10	20:27:27	CAN	192.168.205.1
2	0					
	3	dkot	2022-05-09	06:47:41	USA	192.168.151.1
62	1					
	4	dkot	2022-05-08	02:00:39	USA	192.168.178.7
1	0					
	5	jrafael	2022-05-11	03:05:59	CANADA	192.168.86.23
2	0					
	6	arutley	2022-05-12	17:00:59	MEXICO	192.168.3.24
	0					
	7	eraab	2022-05-11	01:45:14	CAN	192.168.170.2
43	1					
	8	bisles	2022-05-08	01:30:17	US	192.168.119.1
73	0					
	9	yappiah	2022-05-11	13:47:29	MEX	192.168.59.13
6	1					
	10	jrafael	2022-05-12	09:33:19	CANADA	192.168.228.2
21	0					
	11	sgilmore	2022-05-11	10:16:29	CANADA	192.168.140.8

I started by selecting all data from the `log_in_attempts` table. Then, I used a `WHERE` clause with an `OR` operator to filter my results to output only login attempts that occurred on either 2022-05-09 or 2022-05-08.

Retrieve login attempts outside of Mexico

There's been suspicious activity with login attempts, but the team has determined that this activity didn't originate in Mexico.

```

MariaDB [organization]> SELECT *
FROM log_in_attempts WHERE NOT country LIKE 'MEX%';
+-----+
+-----+-----+-----+-----+-----+-----+
| event_id | username | login_date | login_time | country | ip_address |
| success |
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
| 1 | jrafael | 2022-05-09 | 04:56:27 | CAN | 192.168.243.1 |
40 | 1 |
| 2 | apatel | 2022-05-10 | 20:27:27 | CAN | 192.168.205.1 |
2 | 0 |
| 3 | dkot | 2022-05-09 | 06:47:41 | USA | 192.168.151.1 |
62 | 1 |
| 4 | dkot | 2022-05-08 | 02:00:39 | USA | 192.168.178.7 |
1 | 0 |
| 5 | jrafael | 2022-05-11 | 03:05:59 | CANADA | 192.168.86.23 |
2 | 0 |
| 7 | eraab | 2022-05-11 | 01:45:14 | CAN | 192.168.170.2 |
43 | 1 |
| 8 | bisles | 2022-05-08 | 01:30:17 | US | 192.168.119.1 |
73 | 0 |
| 10 | jrafael | 2022-05-12 | 09:33:19 | CANADA | 192.168.228.2 |
21 | 0 |
| 11 | sgilmore | 2022-05-11 | 10:16:29 | CANADA | 192.168.140.8 |
1 | 0 |
| 12 | dkot | 2022-05-08 | 09:11:34 | USA | 192.168.100.1 |

```

I used NOT and LIKE filters. Because of attempts not originating in Mexico, we should first get the attempted country started with MEX then we should return the results which don't start with the MEX. The percentage (%) symbol returns the string that starts with the MEX (Like MEXICO-109, MEXICO-123 ,etc.)

Retrieve employees in Marketing

My team wants to perform security updates on Marketing department and office which starts with 'East-' (like East-170, East-320, etc.)

```

MariaDB [organization]> SELECT * FROM employees WHERE department = 'Marketing' AND office LIKE 'East%';
+-----+-----+-----+-----+-----+
| employee_id | device_id | username | department | office |
+-----+-----+-----+-----+-----+
| 1000 | a320b137c219 | elarson | Marketing | East-170 |
| 1052 | a192b174c940 | jdarosa | Marketing | East-195 |
| 1075 | x573y883z772 | fbautist | Marketing | East-267 |
| 1088 | k865l965m233 | rgosh | Marketing | East-157 |
| 1103 | NULL | randers | Marketing | East-460 |
| 1156 | a184b775c707 | dellery | Marketing | East-417 |
| 1163 | h679i515j339 | cwilliam | Marketing | East-216 |
+-----+-----+-----+-----+-----+
7 rows in set (0.132 sec)

```

First I filtered the department which employees are in the Marketing department, then used the LIKE with the office on WHERE to return that office starts with 'East'.

Retrieve employees in Finance or Sales

In that example, my team now needs to perform different security updates on machines for employees in the Sales and Finance departments. So that, I must return all employees who are working on the department of Sales or Finance.

```
MariaDB [organization]> SELECT * FROM
employees WHERE department = 'Sales' OR 'Finance';
```

employee_id	device_id	username	department	office
1009	NULL	lrodriqu	Sales	South-134
1011	l748m120n401	drosas	Sales	South-292
1024	y976z753a267	iuduike	Sales	South-215
1025	z381a365b233	jhill	Sales	North-115
1035	j236k303l245	bisles	Sales	South-171
1039	n253o917p623	cjackson	Sales	East-378
1041	p929q222r778	cgriffin	Sales	North-208
1057	f370g535h632	mscott	Sales	South-270
1063	l686m140n569	lpope	Sales	East-226
1066	o678p794q957	ttyrell	Sales	Central-444
1071	t244u829v723	zdutchma	Sales	West-348
1072	u905v920w694	esmith	Sales	East-421
1078	a667b270c984	sharley	Sales	North-418
1085	h339i498j269	cperez	Sales	East-325
1086	i281j129k749	lmajumda	Sales	West-499
1089	l358m929n154	jpark2	Sales	West-251
1091	n378o313p469	rtran	Sales	Central-230
1092	o391p779q935	lpark	Sales	West-227
1098	u671v146w618	tarchamb	Sales	North-423
1107	d168e758f876	akajwara	Sales	North-471
1109	f229g533h679	nlocklea	Sales	East-196
1110	g567h376i314	pchaudhu	Sales	Central-428
1111	h835i179j862	jlee	Sales	West-309
1116	m272n572o874	nzhao	Sales	South-100

I used OR to return employees who work in the Sales or Finance department.

Retrieve all employees not in IT

My team needs to make one more security update on employees who are not in the Information Technology department. To make the update, I first have to get information on these employees.

```
MariaDB [organization]> SELECT * FROM employees WHERE NOT department = 'Information Technology';
```

employee_id	device_id	username	department	office
1000	a320b137c219	elarson	Marketing	East-170
1001	b239c825d303	bmoreno	Marketing	Central-276
1002	c116d593e558	tshah	Human Resources	North-434
1003	d394e816f943	sgilmore	Finance	South-153
1004	e218f877g788	eraab	Human Resources	South-127
1005	f551g340h864	gesparza	Human Resources	South-366
1007	h174i497j413	wjaffrey	Finance	North-406
1008	i858j583k571	abernard	Finance	South-170
1009	NULL	lrodriqu	Sales	South-134
1010	k242l212m542	jlansky	Finance	South-109
1011	l748m120n401	drosas	Sales	South-292
1015	p611q262r945	jsoto	Finance	North-271
1016	q793r736s288	sbaelish	Human Resources	North-229
1017	r550s824t230	jclark	Finance	North-188
1018	s310t540u653	abellmas	Finance	North-403
1020	u899v381w363	arutley	Marketing	South-351
1022	w237x430y567	arusso	Finance	West-465
1024	y976z753a267	iuduike	Sales	South-215
1025	z381a365b233	jhill	Sales	North-115
1026	a998b568c863	apatel	Human Resources	West-320
1027	b806c503d354	mrah	Marketing	West-246
1028	c603d749e374	aestrada	Human Resources	West-121
1029	d336e475f676	ivelasco	Finance	East-156

I should find the employees who works on the Information Technology department than I used the NOT to filter that who are not working on this department.

Summary

I applied filters to SQL queries to get specific information on login attempts and employee machines. I used two different tables, log_in_attempts and employees. I used AND,OR,NOT operators to filter the specific information needed for each task. I also used LIKE and the percentage sign (%) wildcard to filter for patterns.